

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
NETWORK SECURITY (CYS360)
(6TH SEMESTER, 3RD Year) LAB EXPERIMENT # 8

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Score:	_____	Date:	_____	Instructor's Signature:	_____

Lab # 8: To configuring a Firewall as a DHCP client that applies for an IPv4 address to access the Internet

Objective: The objective of this lab is to configure a Firewall as a DHCP Client using IPv4 address.

Duration: 3 Hours

Expected Outcomes: By the end of this lab session, students will be able to:

1. Configure Firewall as a DHCP Server
2. Configure unnumbered Interface

Rubrics

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT
SCORE			

Note: Student performance will be assessed based on this rubric during lab sessions. **Scenario**

This lab provides an example for configuring a FW as a DHCP client that applies for an IPv4 address to access the Internet.

Networking Requirements

[Figure 1](#) shows that a FW functions as an egress gateway and connect PCs in an intranet to the Internet. The network plan is as follows:

- An administrator manually specifies an IPv4 address for each PC on the network segment 10.3.0.0/24.
- An interface with a static IPv4 address connects the FW to the intranet.
- Another interface on the FW that functions as a DHCP client applies for a client IPv4 address and a DNS server IP address from a DHCP server and connects the intranet to the Internet.

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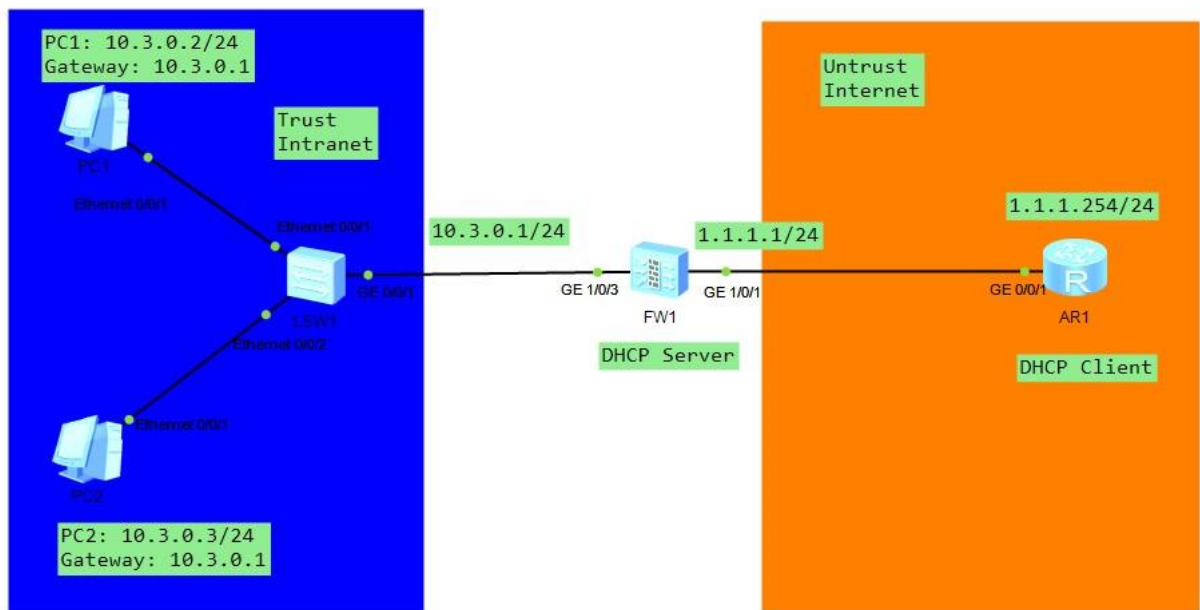


Figure 1 Networking diagram for accessing the Internet using DHCP

Configuration Roadmap

The configuration roadmap is as follows:

1. Enable the DHCP client function on GigabitEthernet 1/0/1 of the FW to obtain a client IPv4 address and a DNS server address from a DHCP server.
2. Specify a static IPv4 address on GigabitEthernet 1/0/3 that connects the FW to the intranet.
3. Configure a security policy and a NAT policy (easy-IP) on the FW.
4. Set the IP addresses of the PCs' gateway and a DNS server to 10.3.0.1. This example provides the configuration procedure on the FW. The configuration procedure for the PCs is not provided.

NOTE:

After the FW obtains an IPv4 address from a DHCP server, the DHCP server issues a default route to the FW that function as a DHCP client. The next hop of the default route is a carrier's device. Therefore, there is no need to configure a default route.

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Procedure

1. Configure the IP address of the interface and assign the interfaces to the security zones.

```
<FW> system-view
[FW] interface GigabitEthernet 1/0/3
[FW-GigabitEthernet1/0/3] ip address 10.3.0.1 24
[FW-GigabitEthernet1/0/3] quit
[FW] firewall zone trust
[FW-zone-trust] add interface GigabitEthernet 1/0/3
[FW] firewall zone untrust
[FW-zone-untrust] add interface GigabitEthernet 1/0/1
[FW-zone-untrust] quit
```

2. Configure the DNS proxy function.

```
[FW] dns proxy enable
[FW] dns resolve
[FW] dns server unnumbered interface GigabitEthernet1/0/1
```

3. Configure GigabitEthernet 1/0/1 as a DHCP client.

```
[FW] interface GigabitEthernet 1/0/1
[FW-GigabitEthernet1/0/1] ip address dhcp-alloc
[FW-GigabitEthernet1/0/1] quit
```

4. Configure a security policy to allow the PCs to access the Internet.

```
[FW] security-policy
[FW-security-policy] rule name policy_sec_1
[FW-security-policy-sec_policy_1] source-address 10.3.0.0 mask 255.255.255.0
[FW-security-policy-sec_policy_1] source-zone trust
[FW-security-policy-sec_policy_1] destination-zone untrust
[FW-security-policy-sec_policy_1] action permit
[FW-security-policy-sec_policy_1] quit
[FW-security-policy] quit
```

5. Configure a NAT policy to translate private network IP addresses into public network IP addresses before PCs access the Internet.

```
[FW] nat-policy
[FW-policy-nat] rule name policy_nat_1
```

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```
[FW-policy-nat-rule-policy_nat_1] source-address 10.3.0.0 mask 255.255.255.0
[FW-policy-nat-rule-policy_nat_1] source-zone trust
[FW-policy-nat-rule-policy_nat_1] egress-interface GigabitEthernet 1/0/1
[FW-policy-nat-rule-policy_nat_1] action source-nat easy-ip
[FW-policy-nat-rule-policy_nat_1] quit
[FW-policy-nat] quit
```

6. Configure ip address as 1.1.1.254/24 on interface g 0/0/1 on AR1(DHCP Client) .

```
[AR1]int g 0/0/1
[AR1-GigabitEthernet0/0/1]ip ad
[AR1-GigabitEthernet0/0/1]ip address 1.1.1.1 24
```

7. Manually assign the ip addresses and gateway on PC1 and PC2 .

Task:

1. Ping from PC1 to the internet on the given ip address 1.1.1.254
2. Ping from PC2 to the internet on the given ip address 1.1.1.254
3. Attach Screenshots

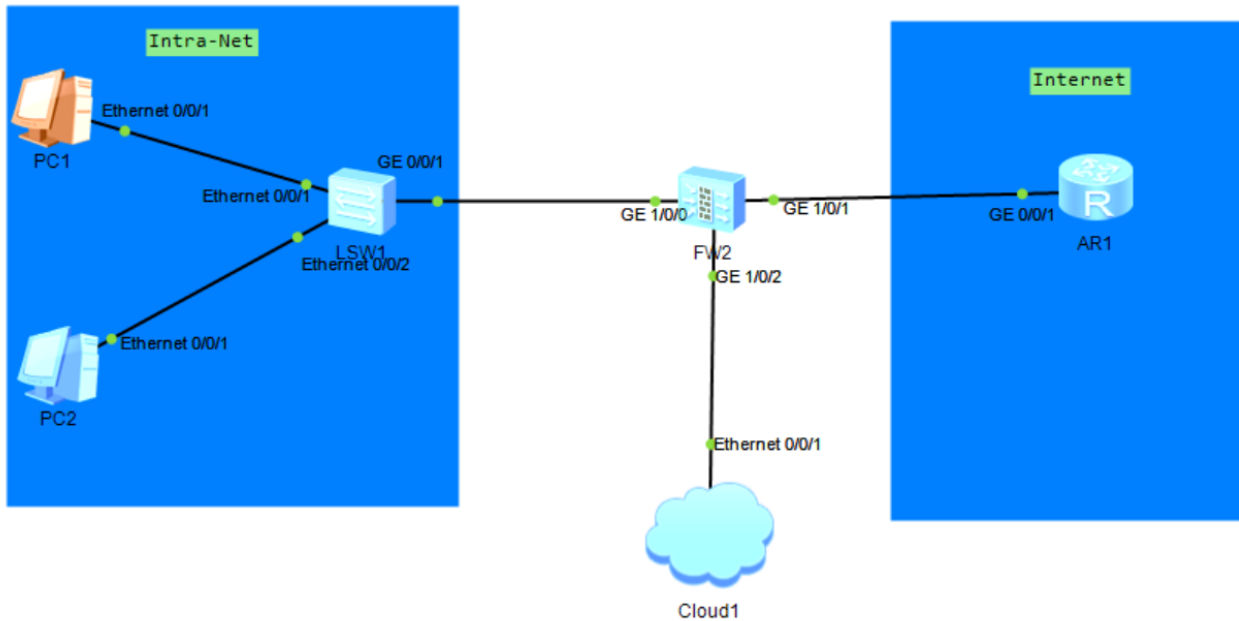
Note: Attach all the screenshots of the configuration. The Firewall name should be your roll number like FW1-22BSCYS001. Upload the .topo and .cfg files on MS Teams Assignment.

Questions:

1. What is an **Unnumbered Interface** and why it is used?

An unnumbered interface is an interface that borrows an IP address from another interface instead of having its own. It is used to save IP addresses and simplify network configuration.

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Interface List

+ Add - Delete

Refresh Interface Name

Interface Name	Zone	IP Address	Connection ...	VLAN/VXL...
GE0/0/0(GE0/METH)	trust(default)	192.168.0.1	Static IP (IPv4)	
GE1/0/0(Intra-net)	trust(public)	10.3.0.1	Static IP (IPv4)	
GE1/0/1(Router)	untrust(public)	1.1.1.254	DHCP (IPv4)	
GE1/0/2	trust(public)	192.168.10.1	Static IP (IPv4)	
GE1/0/3	-- NONE --(public)	---	Static IP (IPv4)	
GE1/0/4	-- NONE --(public)	---	Static IP (IPv4)	
GE1/0/5	-- NONE --(public)	---	Static IP (IPv4)	
GE1/0/6	-- NONE --(public)	---	Static IP (IPv4)	
Virtual-if0	-- NONE --(public)	---	Static IP (IPv4)	

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DNS Server

Transparent DNS Proxy

DNS Server List

+ Add
 ✖ Delete

	DNS Server Address	Obtaining Mode
☐	8.8.8.8	DHCP

+ Add Filter

Index	Name	Descri...	Tag	VLAN ID	Sourc...	Destin...	Sourc...	Destin...	User	Service	Applic...	Sched...	Action	Conte...	Hits	Enable	
☐ 1	Trust-Untrust			any	trust	untrust	10.3.0...	any	any	any	any	any	Permit		5	Reset	☒
2	default	This is...		any	any	any	any	any	any	any	any	any	Deny		29	Re...	☒

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NAT Policy

Source Translation Address Pool

Destination Translation Address Pool

NAT Policy List

</

```
<CR>
[FW-1-23BSCYS006]dis dhcp configuration
2026-02-13 17:05:05.930
  DHCP global running information :
  DHCP                               : Enable
[FW-1-23BSCYS006]dis dhcp cli
[FW-1-23BSCYS006]dis dhcp client
2026-02-13 17:05:18.920
DHCP client lease information on interface GigabitEthernet1/0/1 :
  Current machine state           : Bound
  Internet address assigned via    : DHCP
  Physical address                 : 00e0-fc55-2397
  IP address                      : 1.1.1.254
  Subnet mask                     : 255.255.255.0
  Gateway ip address              : 1.1.1.1
  DHCP server                     : 1.1.1.1
  Lease obtained at                : 2026-02-13 16:56:08
  Lease expires at                : 2026-02-14 16:56:08
  Lease renews at                 : 2026-02-14 04:56:08
  Lease rebinds at                : 2026-02-14 13:56:08
  DNS                             : 8.8.8.8
  Request option list              : 1 3 6 15 28 33 44 121 184
  Class identifier                 : huawei USG6000V1-ENSP
[FW-1-23BSCYS006]
```